

TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (BIOLOGY)

CHAPTER 2: TISSUES

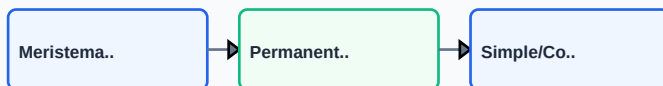
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Biology)
Chapter Name:	Tissues

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TOPIC ROADMAP: PLANT AND ANIMAL TISSUES TAXONOMY



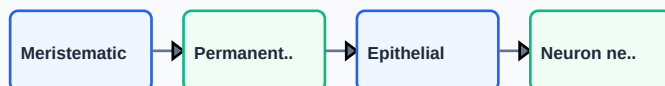
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Tissue	A group of cells similar in structure and origin that work together to perform a specific physiological function.
Meristematic Tissue	Plant tissue consisting of actively dividing cells, responsible for primary and secondary growth, located at growing tips.
Parenchyma	A simple permanent plant tissue consisting of thin-walled living cells, serving functions of storage and photosynthesis.
Sclerenchyma	A simple permanent plant tissue consisting of dead cells with highly thickened lignified walls, providing rigidity and strength.
Xylem	A complex permanent vascular tissue that conducts water and minerals upward from roots to leaves.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

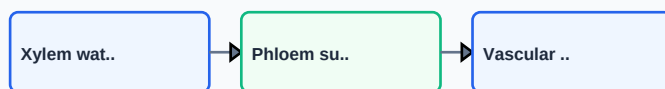
PLANT AND ANIMAL TISSUES MAPPING



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: VASCULAR TISSUE TRANSPORT RATE FACTORS



Essential Formulas, Laws & Identities:

- Complex tissues: Xylem (tracheids, vessels, xylem parenchyma, xylem fibers) and Phloem (sieve tubes, companion cells, phloem parenchyma, phloem fibers).
- Animal tissue classification: Epithelial, Connective, Muscular, Nervous.

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: What is the function of meristematic tissues and where are they located?

Step-by-step Solution:

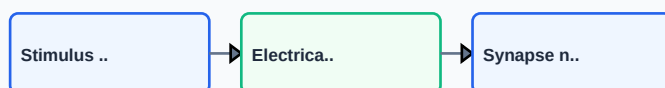
Meristems undergo continuous cell division to grow the plant. They are located at the root and shoot tips (apical meristems), leaf bases (intercalary meristems), and stems girth (lateral meristems).

Example 2: Explain the function of nervous tissue and its cellular unit.

Step-by-step Solution:

Nervous tissue conducts electrochemical nerve impulses to coordinate body responses. Its unit cell is the Neuron, consisting of a cyton, dendrites, and an axon.

WORKED: NERVE IMPULSE CONDUCTION PATHWAY



HOTS (Higher Order Thinking Skills) Challenge

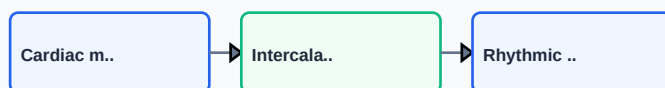
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: How does the structure of Sclerenchyma cells relate to their function in plants?

Analytical Solution Strategy:

Sclerenchyma cells have thick cell walls reinforced with lignin, which makes them impermeable to water and extremely stiff. They lose their cytoplasm at maturity (dead cells) to form hollow, rigid fibers. This provides optimal mechanical support to hulls, seed coats, and leaves.

HOTS: CARDIAC MUSCLE FATIGUE RESISTANCE MECHANISM



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Classify plant tissues in a detailed flowchart.

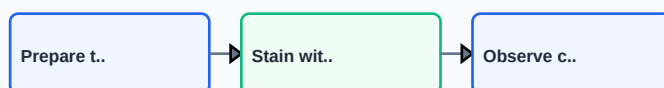
Q2 (2-Mark): Differentiate between Xylem and Phloem in tabular form.

Q3 (2-Mark): Describe the structure and location of three types of muscle fibers: striated, smooth, and cardiac.

Q4 (5-Mark): Explain the components and functions of connective tissue (blood, bone, cartilage).

Q5 (5-Mark): Explain the difference between simple permanent and complex permanent plant tissues.

PRACTICE: HISTOLOGY SLIDE IDENTIFICATION WORKSHEET



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Horticulture and Forestry

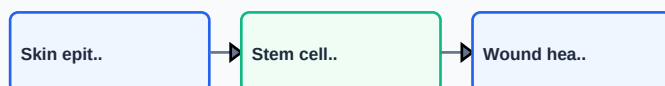
Propagating lateral meristems (grafting cambium) is crucial to cultivate commercial timber and orchards.

Application Case: Medical Skin Grafting

Epithelial tissue culture allows clinics to grow skin sheets from donor cells to treat burn victims.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: TISSUE ENGINEERING IN REGENERATIVE MEDICINE



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm³, N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: PLANT AND ANIMAL TISSUES HIERARCHY

